

Module -1

DC Circuit Analysis and Theorems

Dr. Sathya P
Associate Professor
SENSE

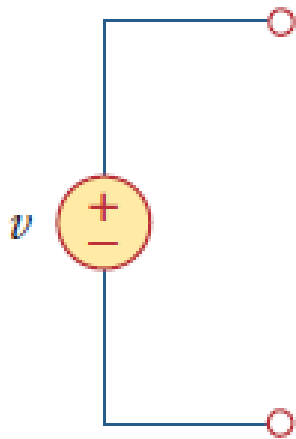
Circuit Elements

- **Element:** An element is the basic building block of a circuit.
- **Circuit:** An electric circuit is an interconnection of the elements.
- **Circuit Analysis:** It is the process of determining voltages across (currents through) the elements of the circuit.
- **Types of elements:** Active and Passive elements.
- **Active elements:** Capable of generating energy.
Eg.:Generators, batteries, electronic devices and operational amplifiers.
- **Passive elements:** Does not generate energy.
Eg.: Resistors, Capacitors and Inductors.

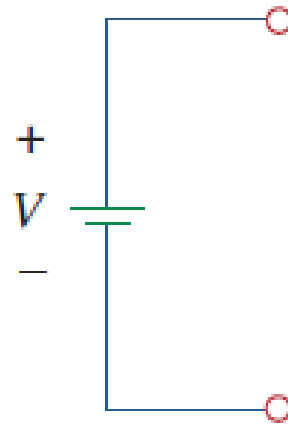
Sources

- Voltage and Current sources are the active elements that generally deliver power to the circuit.
- Types: Independent and Dependent sources

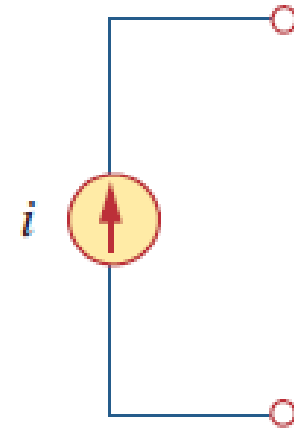
An **ideal independent source** is an active element that provides a specified voltage or current that is completely independent of other circuit elements.



Constant or Time varying voltage



Constant DC voltage

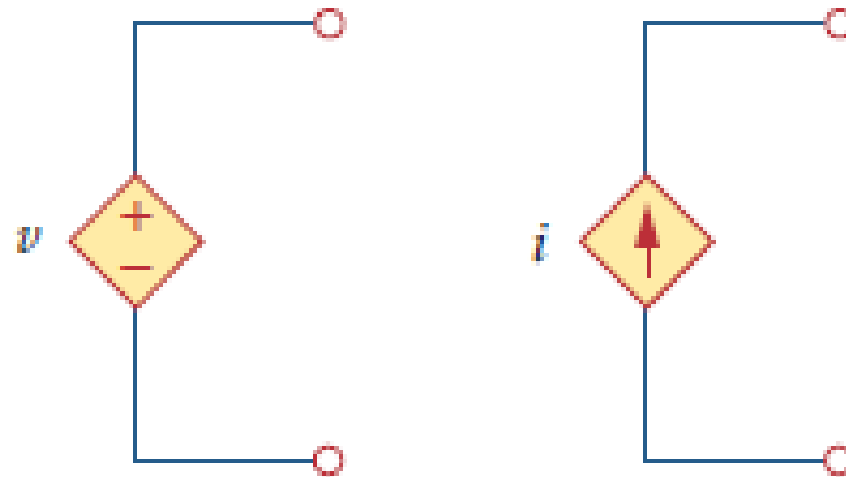


Current source

Dependent Sources

An **ideal dependent** (or **controlled**) **source** is an active element in which the source quantity is controlled by another voltage or current.

1. A voltage-controlled voltage source (VCVS).
2. A current-controlled voltage source (CCVS).
3. A voltage-controlled current source (VCCS).
4. A current-controlled current source (CCCS).



Ohm's Law

Ohm's law states that the voltage v across a resistor is directly proportional to the current i flowing through the resistor.

$$v \propto i$$

$$v = iR$$

The *resistance* R of an element denotes its ability to resist the flow of electric current; it is measured in ohms (Ω).

$$R = \frac{v}{i}$$

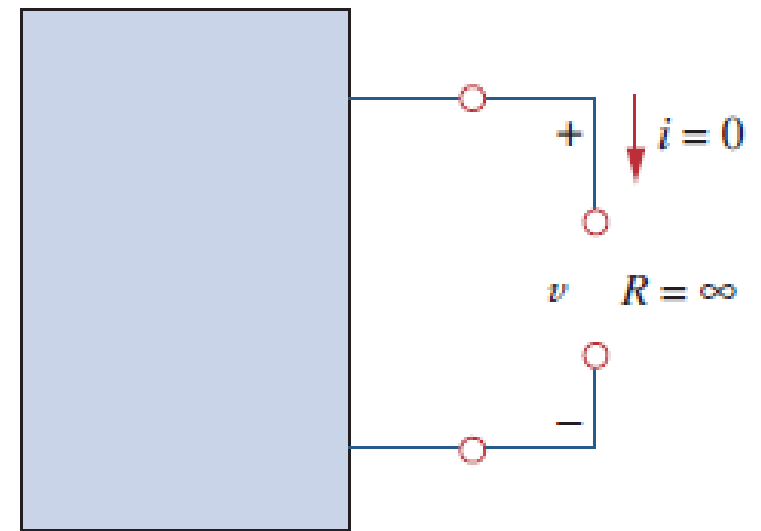
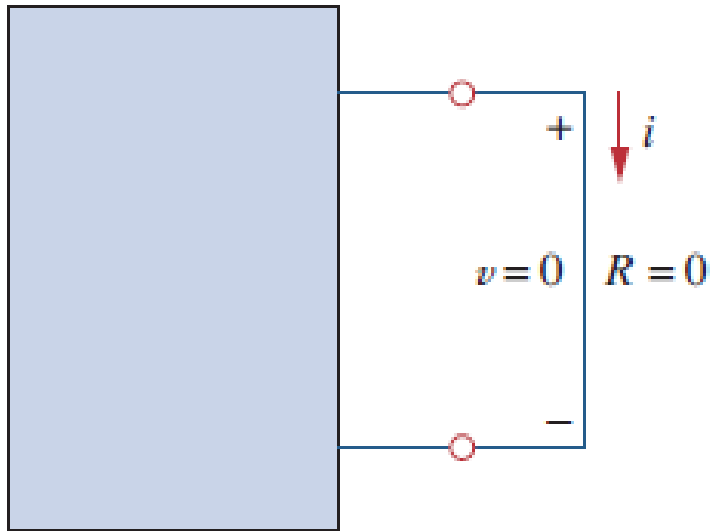
Resistance

A **short circuit** is a circuit element with resistance approaching zero.

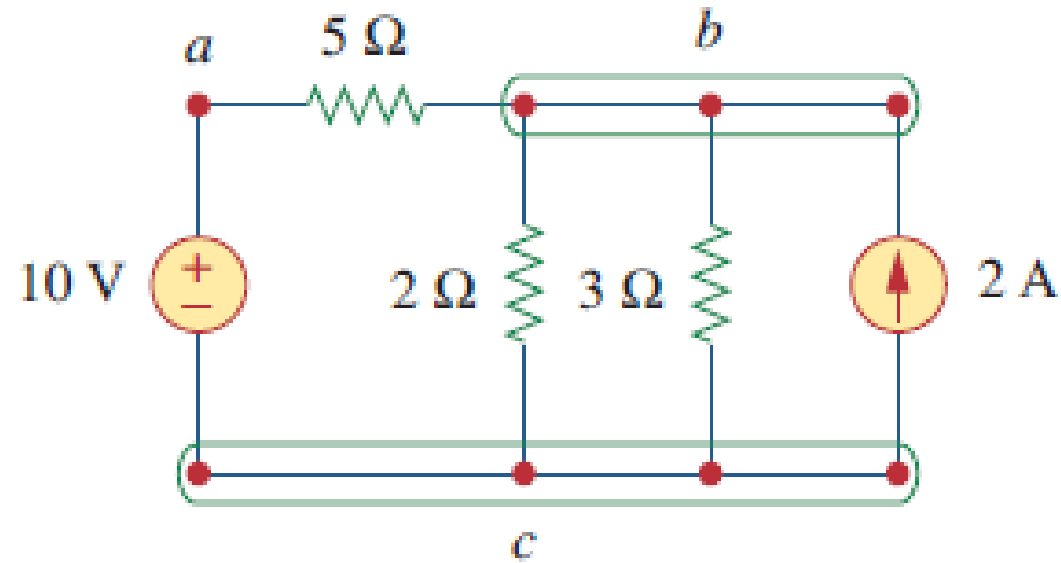
$$v = iR = 0$$

An **open circuit** is a circuit element with resistance approaching infinity.

$$i = \lim_{R \rightarrow \infty} \frac{v}{R} = 0$$



Circuit



A **branch** represents a single element such as a voltage source or a resistor.

A **node** is the point of connection between two or more branches.

Kirchhoff's Current Law (KCL)

Kirchhoff's current law (KCL) states that the algebraic sum of currents entering a node (or a closed boundary) is zero.

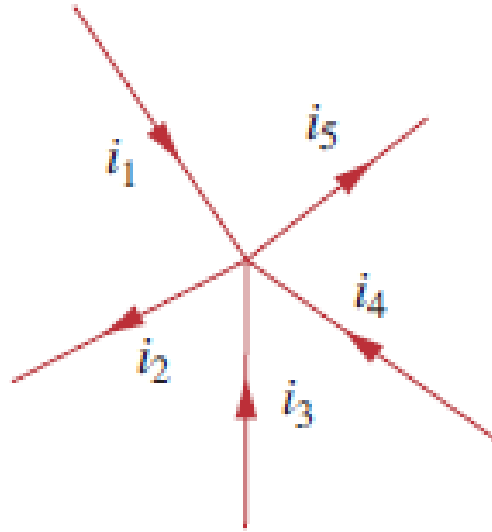
$$\sum_{n=1}^N i_n = 0$$

N is the number of branches connected to the node

i_n is the nth current entering (leaving) the node

The sum of the currents entering a node is equal to the sum of the currents leaving the node.

Kirchhoff's Current Law (KCL)



$$i_1 + (-i_2) + i_3 + i_4 + (-i_5) = 0$$

$$i_1 + i_3 + i_4 = i_2 + i_5$$

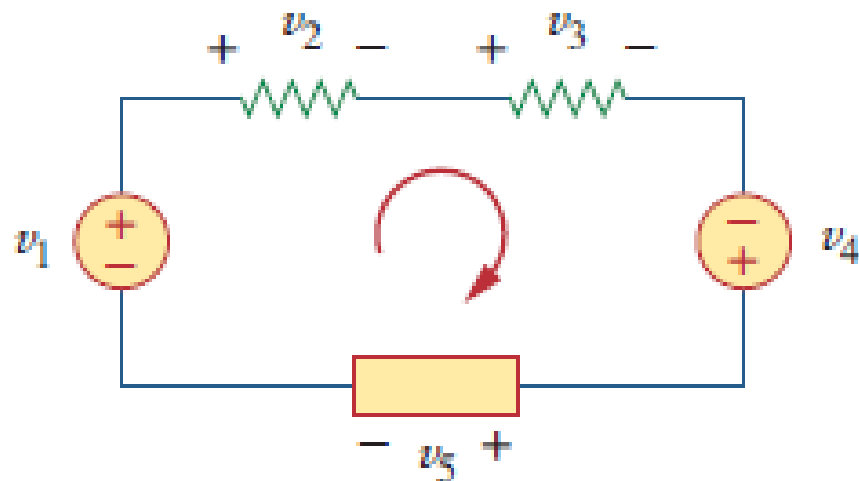
Kirchhoff's Voltage Law (KVL)

Kirchhoff's voltage law (KVL) states that the algebraic sum of all voltages around a closed path (or loop) is zero.

$$\sum_{m=1}^M v_m = 0$$

where M is the number of voltages in the loop (or the number of branches in the loop) and v_m is the m th voltage.

Sum of voltage drops = Sum of voltage rises



$$-v_1 + v_2 + v_3 - v_4 + v_5 = 0$$

$$v_2 + v_3 + v_5 = v_1 + v_4$$

Tutorial -1

- Find the voltages V_1 and V_2 .

Solution:

$$v_1 = 2i, \quad v_2 = -3i$$

$$-20 + v_1 - v_2 = 0$$

$$-20 + 2i + 3i = 0$$

$$5i = 20 \quad \Rightarrow \quad i = 4 \text{ A}$$

$$v_1 = 8 \text{ V}, \quad v_2 = -12 \text{ V}$$

